Enhanced Entity Relationship (EER) Modeling

CSE 341
Introduction to Database Systems
Winter 2024
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Introduction

- In previous lecture, we discussed the basic concepts of the ER model.

- These basic concepts are normally adequate for building data models of traditional, administrative-based database systems such as stock control, product ordering, and customer invoicing

Enhanced Entity-Relationship Model

- Since the1980s, there has been an increase in emergence of new database applications with more demanding requirements.

- Basic concepts of ER modelling are not sufficient to represent requirements of newer, more complex applications.

- Response is development of additional ‘semantic’ modelling concepts.

The Enhanced Entity-Relationship Model

- Semantic concepts are incorporated into the original ER model and called the Enhanced Entity-Relationship (EER) model.

- Examples of additional concept of EER model is called specialization / generalization.

Specialization / Generalization

- The concept of specialization/generalization is associated with special types of entities known as superclasses and subclasses, and the process of attribute inheritance.

- We will also describe the two main types of constraints on superclass/subclass relationships called participation and disjoint constraints.

Specialization / Generalization

- Superclass
  - An entity type that includes one or more distinct subgroupings of its occurrences.
- Subclass
  - A distinct subgrouping of occurrences of an entity type.

Specialization / Generalization

- Superclass/subclass relationship is one-to-one (1:1).
- Superclass may contain overlapping or distinct subclasses.
- Not all members of a superclass need be a member of a subclass.

Specialization / Generalization

- Attribute Inheritance
  - An entity in a subclass represents same ‘real world’ object as in superclass, and may possess subclass-specific attributes, as well as those associated with the superclass.

Specialization / Generalization

- Specialization
  - Process of maximizing differences between members of an entity by identifying their distinguishing characteristics.

- Generalization
  - Process of minimizing differences between entities by identifying their common characteristics.

AllStaff table holding details of all staff

Attributes appropriate for all staff

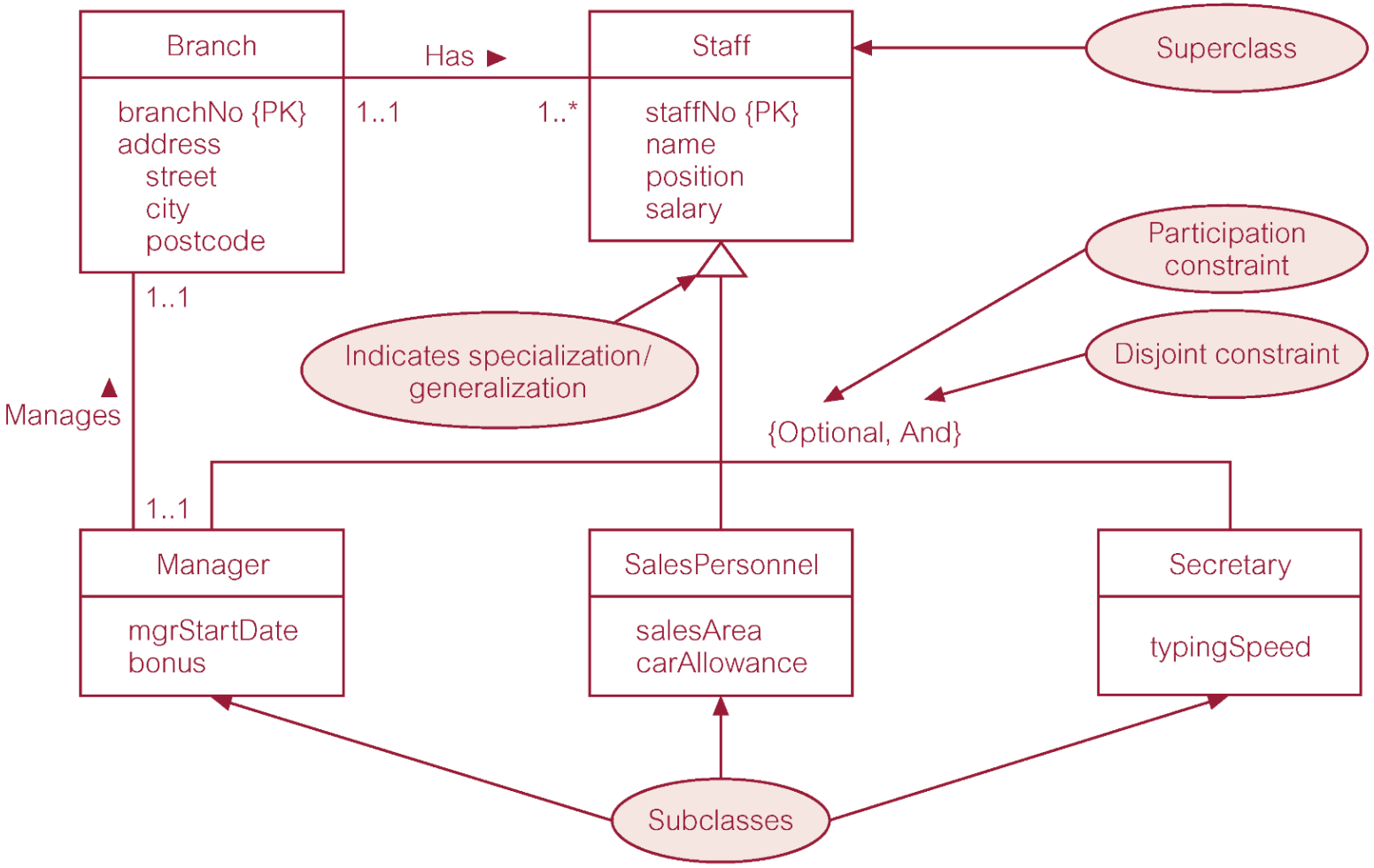
Attributes appropriate for branch Managers

Attributes appropriate for Sales Personnel

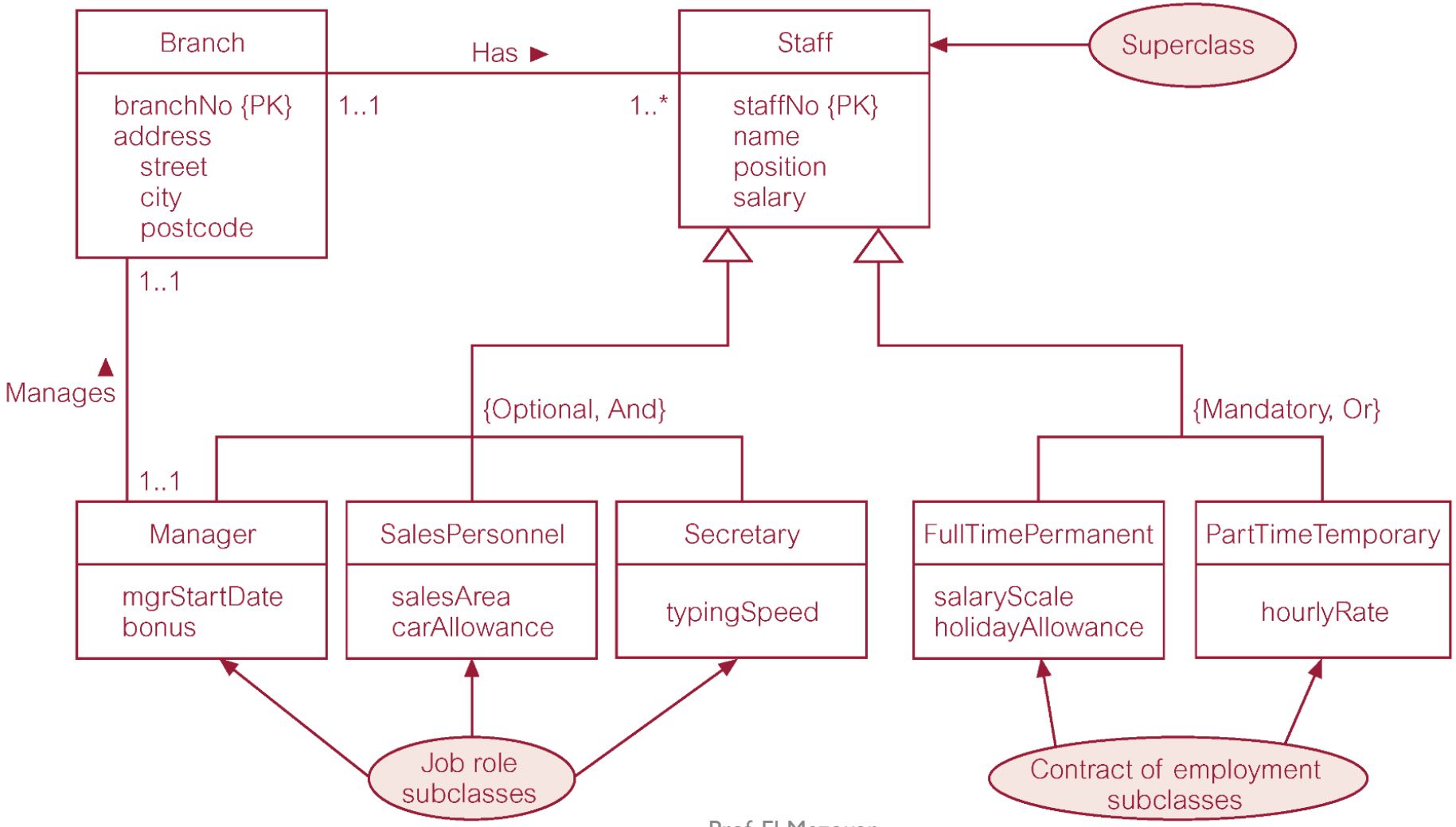
Attribute appropriate for Secretarial staff

staffNo	name	position	salary	mgrStartDate	bonus	sales Area	car Allowance	typing Speed
SL21	John White	Manager	30000	01/02/95	2000			
SG37	Ann Beech	Assistant	12000					
SG66	Mary Martinez	Sales Manager	27000			SA1A	5000	
SA9	Mary Howe	Assistant	9000					
SL89	Stuart Stern	Secretary	8500					100
SL31	Robert Chin	Snr Sales Asst	17000			SA2B	3700	
SG5	Susan Brand	Manager	24000	01/06/91	2350			

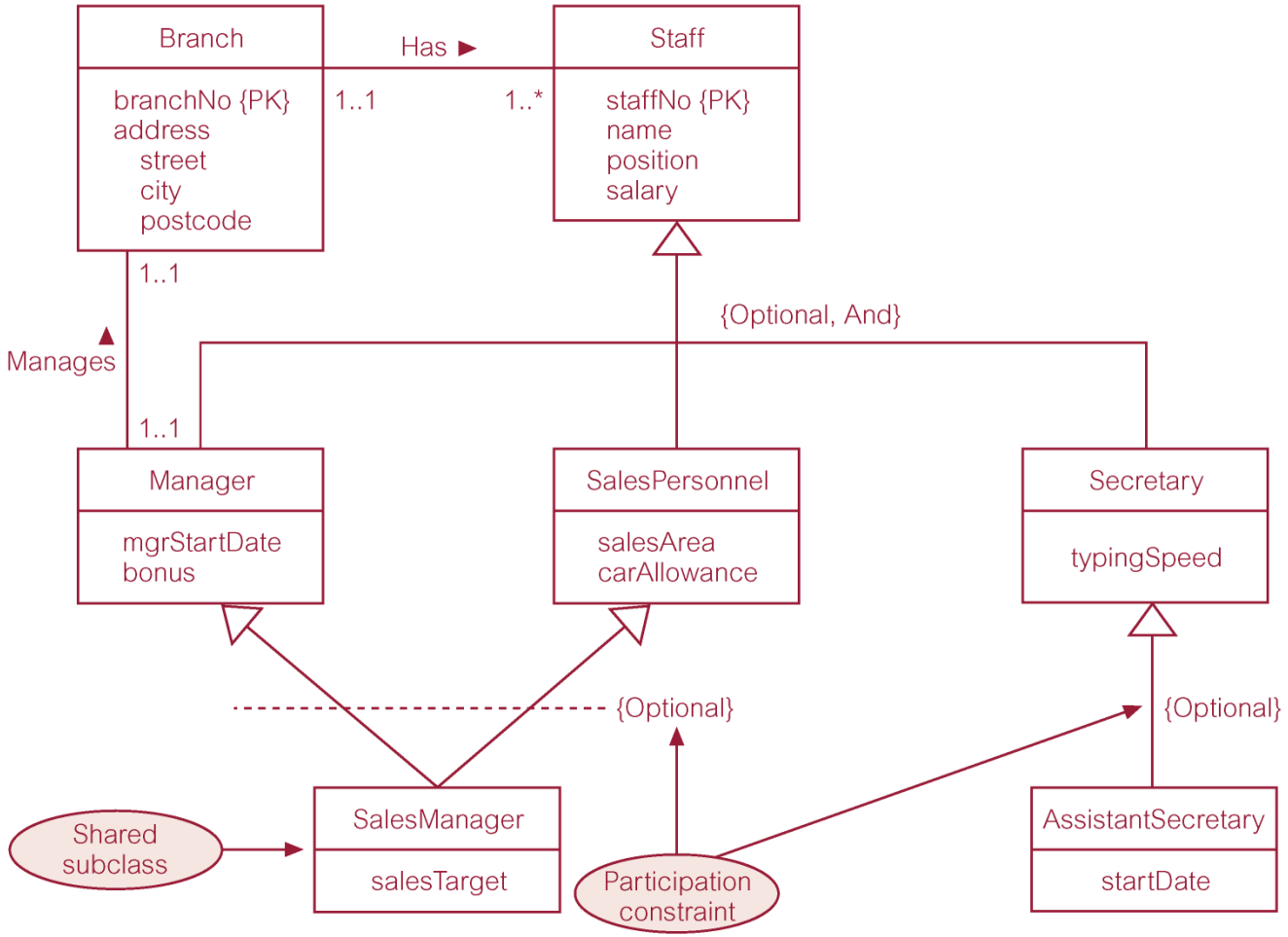
Specialization/generalization of Staff entity into subclasses representing job roles



Specialization/generalization of *Staff* entity into job roles and contracts of employment



EER diagram with shared subclass and subclass with its own subclass



Constraints on Specialization / Generalization

- Two constraints that may apply to a specialization/generalization:
  - participation constraints
  - disjoint constraints.

- Participation constraint
  - Determines whether every member in superclass must participate as a member of a subclass.
  - May be *mandatory* or *optional*.

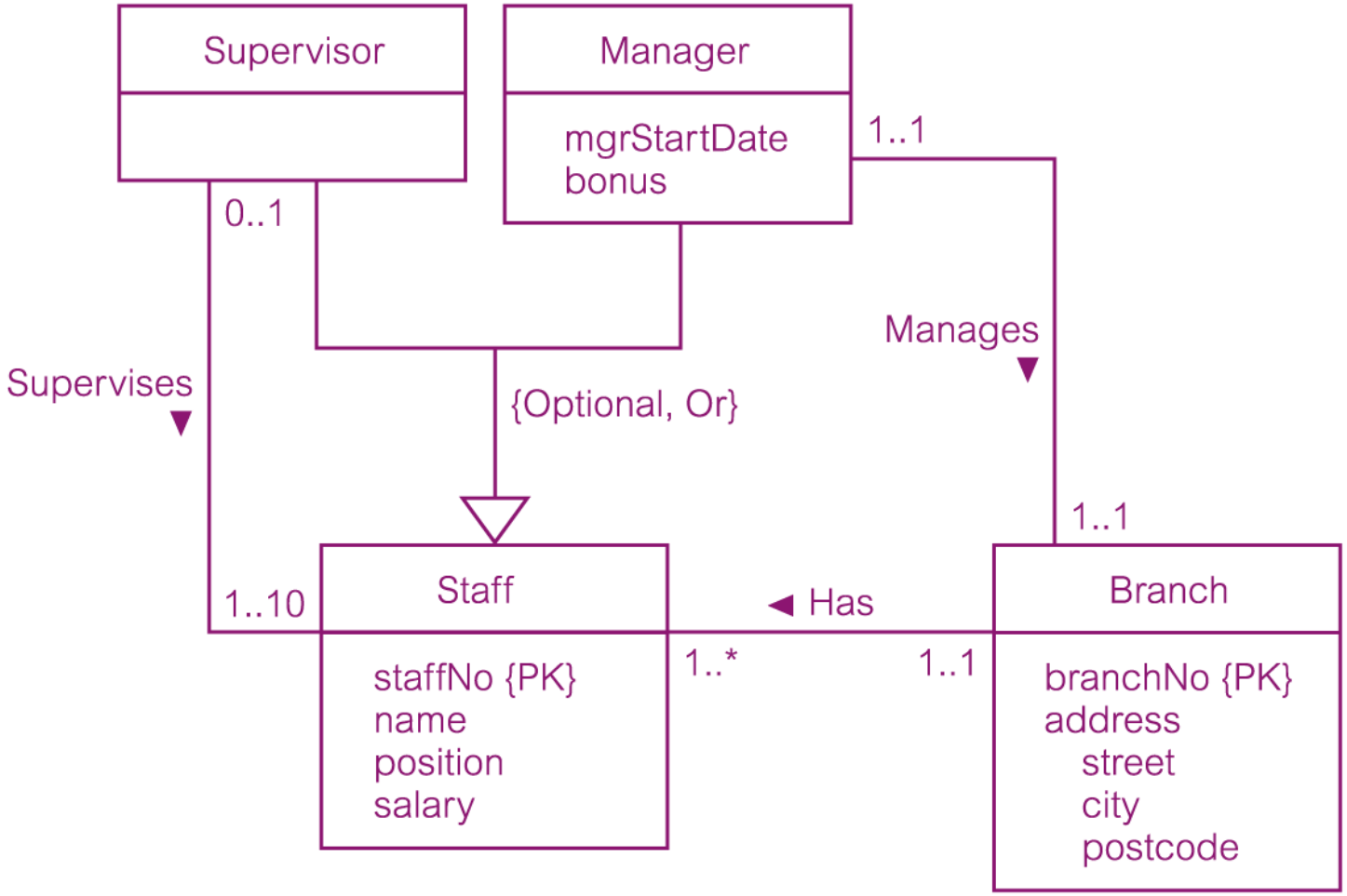
Constraints on Specialization / Generalization

- Disjoint constraint
  - Describes relationship between members of the subclasses and indicates whether member of a superclass can be a member of one, or more than one, subclass.
  - May be *disjoint* or *nondisjoint*.

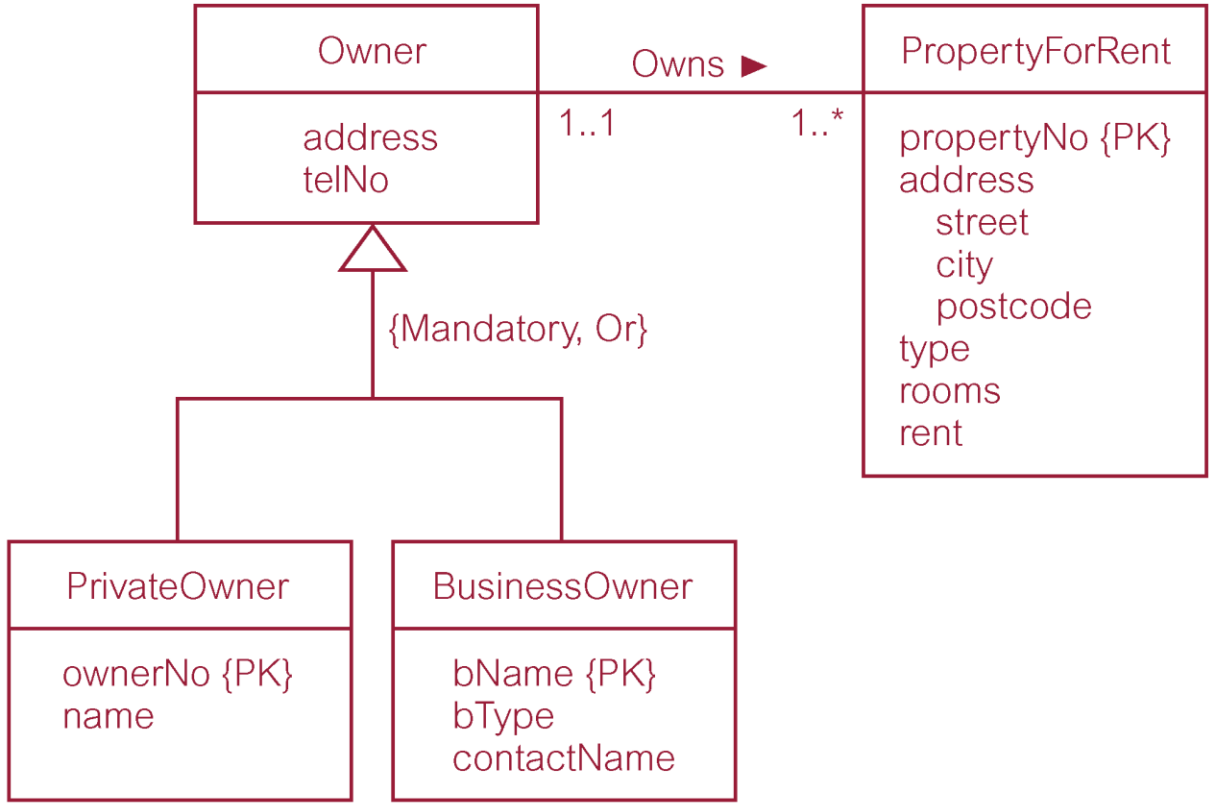
Constraints on Specialization / Generalization

- There are four categories of constraints of specialization and generalization:
  - mandatory and disjoint
  - optional and disjoint
  - mandatory and nondisjoint
  - optional and nondisjoint.

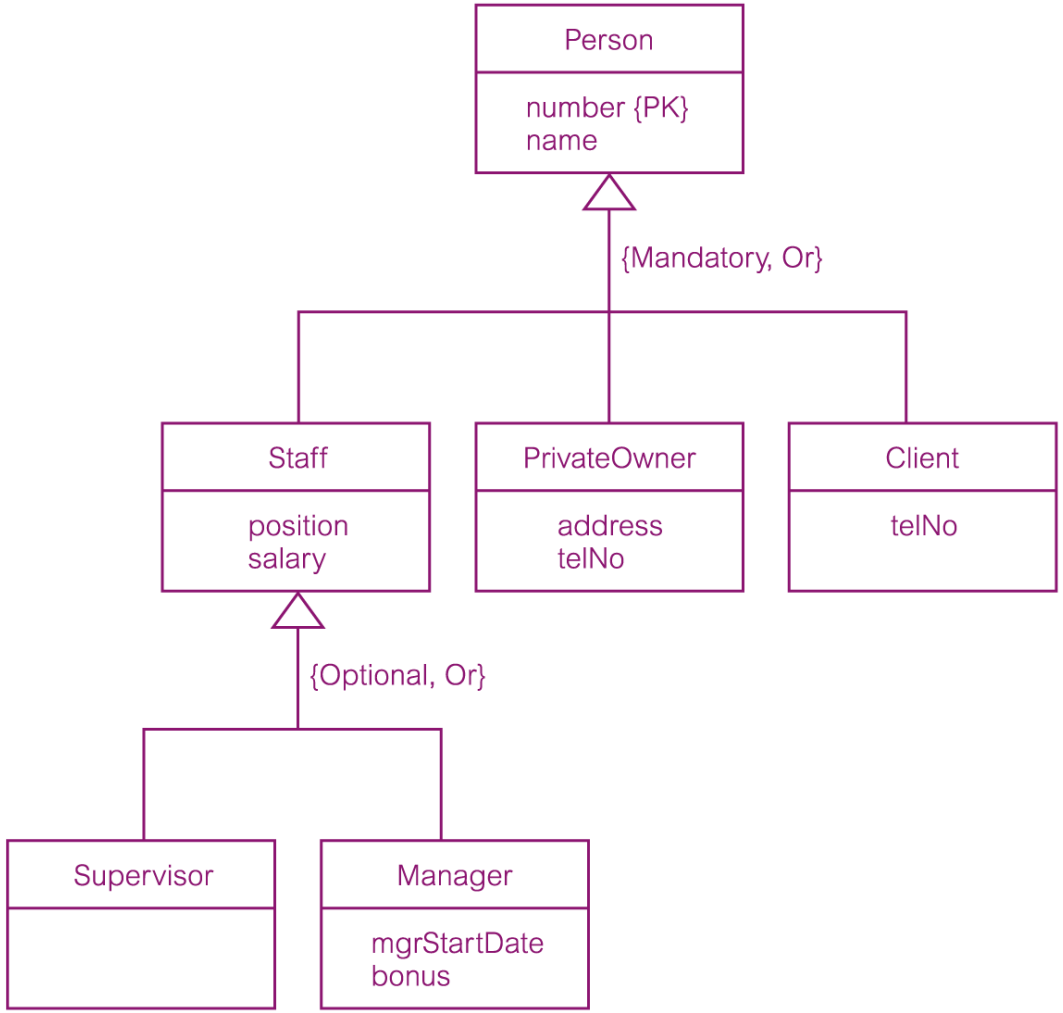
DreamHome worked example - Staff Superclass with Supervisor and Manager subclasses



DreamHome worked example - Owner *Superclass* with *PrivateOwner* and *BusinessOwner* subclasses



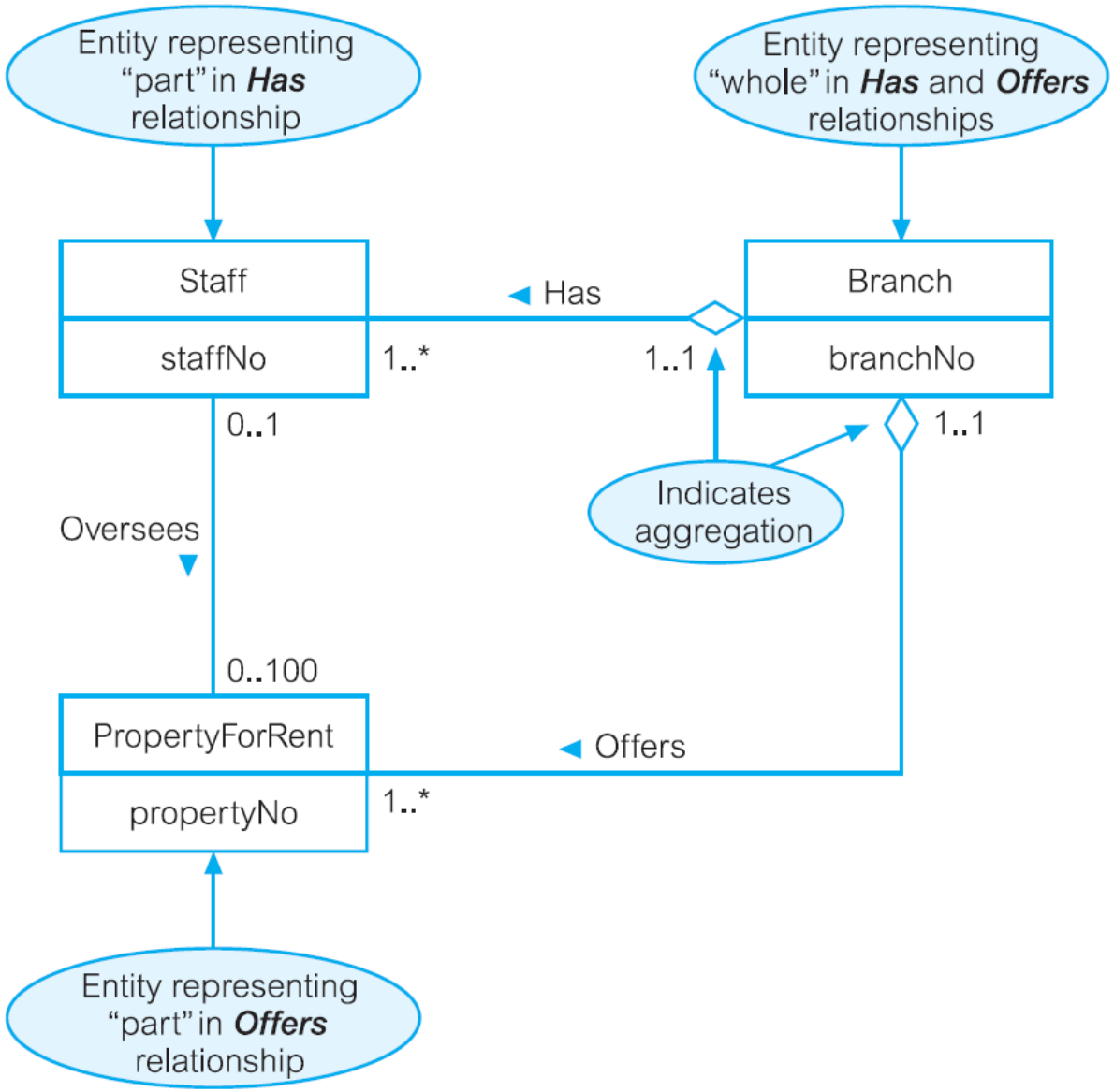
DreamHome worked example - *Person* superclass with *Staff*, *PrivateOwner*, and *Client* subclasses



Aggregation

Represents a "has-a" or "is-part-of" relationship between entity types, where one represents the "whole" and the other the "part."

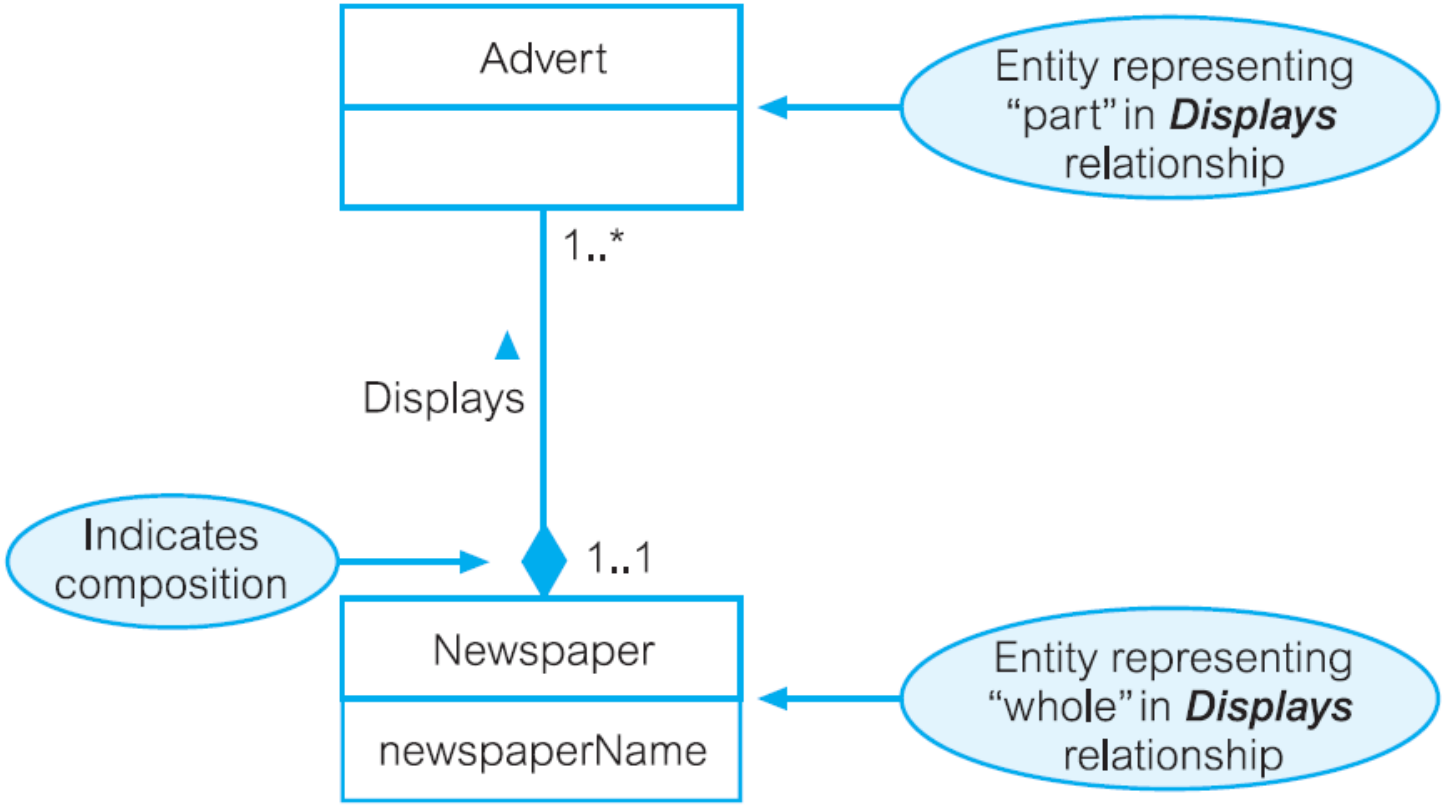
Aggregation



Composition

A specific form of aggregation that represents an association between entities, where there is a strong ownership and coincidental lifetime between the "whole" and the "part."

Composition



Next…

Relational Model